SERIES

BLAKE FARMAN

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Name: Solutions

Theorem. *The geometric series*

$$\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + \dots$$

converges if $|r| < 1$ *and diverges otherwise. The sum of the convergent series is*

$$s = \frac{a}{1-r}, \ |r| < 1.$$

Observation: For a geometric series, we can always find the value of r by taking the ratio of any two consecutive terms:

$$\frac{a_{n+1}}{a_n} = \frac{ar^n}{ar^{n-1}} = r.$$

1. Determine whether the series

$$4 + 3 + \frac{9}{4} + \frac{27}{16} + \dots$$

converges or diverges. If it is convergent, find its sum.

$$4 + 3 + \frac{9}{4} + \frac{27}{16} + \dots = 4 + 4\left(\frac{3}{4}\right) + 4\left(\frac{3}{4}\right)^2 + 4\left(\frac{3}{4}\right)^3$$

$$= \sum_{n=1}^{\infty} 4\left(\frac{3}{4}\right)^{n-1}$$

$$= \frac{4}{1 - 3/4}$$

$$= \frac{4}{\left(\frac{1}{4}\right)}$$

$$= \boxed{16}$$

2. Determine whether the series

$$\sum_{k=1}^{\infty} \frac{k(k+14)}{(k+15)^2}$$

converges or diverges. If it converges, find its sum.

$$\lim_{k\to\infty} \frac{k(k+4)}{(k+15)^2} = \lim_{k\to\infty} \frac{k^2+4k}{k^2+30k+15^2} = 1 \neq 0$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{k(k+4)}{(k+15)^2} \text{ diverges by } n^{th} \text{ Term Test for Divergence.}$$

3. Determine whether the series

$$\sum_{n=1}^{\infty} \frac{2^n+1}{3^n}$$

converges or diverges. If it converges, find its sum.

$$\sum_{n=1}^{\infty} \frac{2^n}{3^n} = \sum_{n=1}^{\infty} \frac{2}{3}\left(\frac{2}{3}\right)^{n-1} = \frac{\frac{2}{3}}{1-2/3} = \frac{2}{3}\Big/\left(1/3\right) = 2$$

$$\sum_{n=1}^{\infty} \frac{1}{3^n} = \sum_{n=1}^{\infty} \frac{1}{3}\left(\frac{1}{3}\right)^{n-1} = \frac{\frac{1}{3}}{1-\frac{1}{3}} = \frac{1}{3}\left(\frac{3}{2}\right) = \frac{1}{2}$$

$$\Rightarrow \sum_{n=1}^{\infty} \frac{2^n+1}{3^n} = \sum_{n=1}^{\infty}\left(\frac{2}{3}\right)^n + \sum_{n=1}^{\infty}\left(\frac{1}{3}\right)^n = 2+1/2 = \boxed{\frac{5}{2}}$$

4. Express $0.\bar{8}$ as a rational number (i.e. a ratio of two integers).

$$0.\overline{8} = 0.8 + 0.08 + 0.008 + \cdots = \frac{8}{10} + \frac{8}{10}\left(\frac{1}{10}\right) + \frac{8}{10}\left(\frac{1}{10}\right)^2 + \cdots$$

$$= \sum_{n=1}^{\infty} \frac{8}{10}\left(\frac{1}{10}\right)^{n-1}$$

$$= \frac{\frac{8}{10}}{1 - \frac{1}{10}}$$

$$= \frac{8}{10}\left(\frac{10}{9}\right)$$

$$= \boxed{\frac{8}{9}}$$

5. Express $2.\overline{516}$ as a rational number.

$$.\overline{516} = .516 + .000516 + .000000516 + \cdots$$

$$= \frac{516}{10^3} + \frac{516}{10^6} + \frac{516}{10^9} + \cdots$$

$$= \frac{516}{10^3} + \frac{516}{10^3}\left(\frac{1}{10^3}\right) + \frac{516}{10^3}\left(\frac{1}{10^3}\right)^2 + \cdots$$

$$= \frac{\frac{516}{10^3}}{1 - \frac{1}{1000}}$$

$$= \frac{516}{1000}\left(\frac{1000}{999}\right)$$

$$= \frac{516}{999}$$

$$= \frac{3(172)}{999}$$

$$= \frac{172}{333}$$

$$\Rightarrow 2.\overline{516} = \frac{666}{333} + \frac{172}{333} = \boxed{\frac{838}{333}}$$